

TAHIA ANAN DHIRA

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PROFILE

Applied economist and computational social scientist (MSc Applied Social Data Science, LSE, Distinction; MSS Economics, University of Dhaka) working on the intersection of empirical econometrics, network analysis, and environmental and economic outcomes. Current research applies panel econometrics and network modelling to trade-embodied emissions, migration, and trade relationships, with a peer-reviewed record including PLOS ONE and SSM – Health Systems. Currently Assistant Professor of Economics at the University of Dhaka, seeking to pursue doctoral research in climate and energy finance.

COMPUTATIONAL & QUANTITATIVE SKILLS

- Econometrics: panel-data and time-series methods (fixed/random effects, PCSE, FGLS, GMM, ARDL) selected according to data structure — integration order, cross-sectional dependence; structural gravity/PPML estimation; factor analysis (EFA/CFA), principal component analysis, structural equation modelling
- Programming: STATA · Python (pandas, NumPy, NetworkX, Matplotlib, scikit-learn) · R (igraph, statnet, ergm, tergm, stm, networkDynamic, quanteda)
- Network analysis: ERGM, STERGM, dynamic and weighted networks, centrality, community detection, core-periphery analysis, block modelling, cluster analysis
- Machine learning & NLP: sentiment analysis (VADER), structural topic modelling (STM, BERTopic), applied machine learning; additional training in word embeddings, LDA, Wordfish, and transformer-based text classification (BERT)
- Research workflow: Git/GitHub; Gemini API; reproducible pipelines; data visualization; LaTeX
- Data collection: Qualtrics; YouTube Data API v3

EDUCATION

MSc in Applied Social Data Science (Distinction) — London School of Economics and Political Science (2023)

Coursework: Social Network Analysis; Quantitative Text Analysis; Applied Machine Learning; Managing and Visualizing Data; Research Design

MSS in Economics (CGPA 3.88/4.00) — University of Dhaka (2017)

BSS in Economics (CGPA 3.94/4.00) — University of Dhaka (2016)

CURRENT RESEARCH MANUSCRIPTS

Rethinking the Income–Emissions Relationship: Evidence from Trade-Embodied CO₂ Emissions Networks (*Submitted, 2026*)

- Built reproducible computational workflows integrating large-scale international trade-embodied emissions and macroeconomic datasets
- Constructed weighted global trade-embodied-emissions networks, computed network metrics and country positions
- Combined network analysis with panel econometrics to examine how development and network position shape domestic and trade-embodied CO₂ emissions
- Corresponding author; built reproducible workflows across Python, R, and Git

Product Reproducibility and International Trade Patterns: Evidence from Cultural Goods (*Under review, 2026*)

- Constructed computational pipelines to harmonize and process bilateral trade data across 190 countries
- Combined structural gravity modelling (PPML) with dynamic network modelling (STERGM) to examine trade-relationship formation, persistence, and trade intensity
- Corresponding author; built reproducible workflows across Python, R, and Git

Global Migration Network and the Geography of Development (*Working paper, 2026*)

- Built a 23-year dynamic panel of directed, weighted bilateral migration networks
- Developed reproducible R workflows for network construction, visualization, and panel data analysis
- Contributed to network model development, core-periphery detection, and manuscript preparation

Gendered Cultural Hierarchies in Social Media Audience Discourse: Evidence from Bangladeshi Artists (*Working paper, 2026*)

- Built scalable Python pipelines (YouTube Data API v3) to collect and process over 8 million multilingual comments

- Applied sentiment analysis (VADER) and structural topic modelling (quanteda/STM) to audience discourse across artists, genders, and artistic categories
- Designed a human-in-the-loop classification workflow using the Gemini API, with manual validation across ~48,000 videos

PEER-REVIEWED PUBLICATIONS

- Huque, R., Abdullah, S. M., **Dhira, T. A.**, et al. (2023). Determinants of client satisfaction with health insurance: Evidence using a mixed-method approach from Bangladesh. *SSM - Health Systems*, <https://doi.org/10.1016/J.SSMHS.2025.100058>
- Rahman, M. A., **Dhira, T. A.**, et al. (2022). PHQ-9 validation in Bangladesh. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0269634>
- **Dhira, T. A.**, Rahman, M. A., et al. (2021). GAD-7 validation in Bangladesh. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0261590>
- Mehareen, J., Rahman, M. A., **Dhira, T. A.**, et al. (2021). Mental health during COVID-19. *Journal of Affective Disorders Reports*. <https://doi.org/10.1016/j.jadr.2021.100179>

EARLIER RESEARCH EXPERIENCE

Capstone Project, MSc ASDS, LSE — Friendship Network Formation: Pro-sociality and Personality Type (2023)

- Designed and implemented a behavioural classroom experiment on homophily and friendship network formation
- Estimated Exponential Random Graph Models (ERGMs) to identify structural and attribute-based drivers of network formation

Coursework project, ST445, LSE — Topic-Based Content and Sentiment Analysis of COVID-19 Vaccine Discourse on Reddit (2022–2023)

- Visualized a subreddit-level network in which two subreddits are linked by shared frequent commenters among the 40 most active authors, identifying the most central communities and informing the choice to scope subsequent analysis to the top 20 subreddits
- Contributed to report writing, documentation, and referencing on a project led by a classmate, who collected, processed, and modeled the underlying multi-million-comment corpus

Mental Health of University Students Before and After Online Education (UGC-funded) (2020–2021)

- Contributed to design and implementation of a longitudinal survey; coordinated data collection and survey-instrument validation

Willingness to Pay for Treating Mental vs. Physical Illness (BMRC-funded) (2020–2021)

- Contributed to coordination and implementation of a comparative survey for university students and adults

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Economics, University of Dhaka (October 2025 – present)

Lecturer, Department of Economics, University of Dhaka (November 2018 – September 2025)

- Taught 10 undergraduate economics courses; supervised undergraduate dissertations and research projects
- Trained undergraduate and graduate students in STATA; developed computational course materials in R, Python, and STATA

Lecturer, Department of Economics, East West University (March 2018 – October 2018)

AWARDS

- Commonwealth Master's Scholarship (2022)
- Gold Medals for highest CGPA, BSS & MSS Economics, University of Dhaka (2016, 2017)

LANGUAGES

Advanced fluency in English and Bangla